

Rust direct-rs Review — Consolidated Tutoring + CE Code Review

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1 Overview

This file consolidates two sources into one reference:

1. **Tutoring homework list** — prioritized by rust-skills rule impact (CRITICAL → LOW), derived from a full read of rust/direct-rs/src/*.rs and rust/Cargo.toml.
2. **CE code-review synthesis** — 6 parallel persona reviewers (correctness, testing, maintainability, performance, adversarial, project-standards), each forced through the rust-skills rubric, structured JSON, confidence-gated ≥75, deduped.

Crate: rust/direct-rs as staticlib+cdylib via cxx into the Qt6/CUDA JTML host.

Intent: Verified DIRECT port — Jones POH/convex-hull, trisection, volume tiling, OrderedFloat determinism, batch Cost::eval, 10M-eval Rastrigin reach.

Suite at review: 46 passed / 0 failed / 2 ignored (Rastrigin 10M + budget overshoot parked as #[ignore = "..."]). Your5k → 13-15k it/s is real — lean BTreeMap<OrderedFloat, (OrderedFloat, u64)> + Hyperbox{[u32;6]} vs C++ std::map churn.

Already landed this session (not in TODO): deleted ghost pose.rs + dup_diag.rs, removed ghost trait Cost in direct_data_storage.rs, #[cfg(test)] on Sphere, #[cfg(test)] on bench/=utils, deduped new()→new_with_strat, captured best() once, parked long-run tests, wired AnisotropicSphere + fstar()/xstar() oracle use.

Subagent artifacts: /home/ajj/.pi/agent/sessions/--home-ajj-repo-uf-JTML--/subagent-artifacts/structured-output/<runId>/pi-subagent-*/output.json, workflow 052a3f67-213b-4245-8c08-6f7979b1367f (raw 47 → merged 47 findings).

2 Tutoring Homework List

How to use: Work top-down. Tier 1 is contract/interface — do next session. Tier 2 is quality/hardening — do soon. Tier 3 is advisory — pick when touched.

2.1 Tier 1 — Should do next session (contract / interface)

2.1.1 1. Remove println! timing — obs-tracing-over-log, obs-library-facade

Delete the println!("{:?} per iteration") block. Library crates emit via tracing, never install a subscriber. Fold stats into RunOutcome (avg_per_call_us, it_per_sec) or gate behind #[cfg(debug_assertions)]

2.1.2 2. Replace INFINITY sentinel — type-option-nullable, api-must-use

Return `Result<RunOutcome, NoFiniteIncumbent>` or `best_cost: Option<f64>`. Add `#[must_use]` on `run()`. Sentinels are indistinguishable from real $+\infty$.

2.1.3 3. Add `From<[f64;6]>` for Pose — conv-from-not-into, type-newtype-ids

impl `From<[f64;6]>` for `Pose (+inverse)`. Two hand-rolled index blocks → two one-liners, never transpose an axis.

2.1.4 4. Batch the FFI bridge — wall-clock knob

Per-pose `UniquePtr<Point6D>` + one `cxx` call per pose defeats batch GPU eval. Add `evaluate_batch(&CppCost, &[f64], &mut [f64])` seam; keep per-pose fallback for tests.

2.1.5 5. Fix budget overshoot

Guard fires before $2 * |\text{POH}|$ batch. Cap with `remaining = budget.saturating_sub(calls)` and `min(|POH|, ceil(remaining/2))` then re-enable the property test.

2.2 Tier 2 — Worth doing soon (quality / hardening)

2.2.1 6. Rename Direction variants — name-types-camel, allow_{attributes} ban

`#[allow(non_camel_case_types)]` violates your own `allow_attributes_without_reason = "warn"`. Rename to `X, Y, Z, Xa, Ya, Za` or `Axis::X`, drop the attribute.

2.2.2 7. Use integer exponents and cache size — opt-inline-small

`powf` → `powi(-(2*d as i32))` (integer exponents) + consider caching size on box. Preserve bit-exact term ordering (`size_is_bit_identical_under_depth_permutation` test).

2.2.3 8. Reuse Vec buffers — mem-with-capacity, mem-reuse-collections

`Vec::with_capacity` + reusable scratch buffers as optimizer fields for allocation-free inner loop.

2.2.4 9. Document public API — doc-all-public, doc-errors-section

No `///` on public items. Document determinism + budget contract, `POHSettings` variants, whether discriminants `=0/=1` are FFI-stable.

2.2.5 10. Pin MSRV — proj-msrv-declare

No `rust-version`. Pin `rust-version = "1.85"` (2024 edition floor).

2.2.6 11. Tune release profile — `perf-release-profile`, `opt-lto-release`, `opt-codegen-units`

No `[profile.release]`. For `staticlib` into Qt/CUDA: `lto="thin"` `codegen-units=1` is free wall-clock.

2.2.7 12. Add criterion bench — `test-criterion-bench`, `perf-black-box-bench`

File is `fixtures/tests`, not a harness. Add `criterion+black_box` micro-bench of `DirectOptimizer::run` vs trivial Cost to substantiate 5k→13k claims.

2.3 Tier 3 — Advisory / food-for-thought

2.3.1 13. Jones ϵ -condition (algorithmic)

POH lacks classic ϵ -filter (`skip if improvement <  $\epsilon \cdot \max(|f_{\min}|, 1e-8)$`). Biggest lever for Ackley/Rastrigin — likely cuts 10M substantially.

2.3.2 14. Float-order fragility in `size()`

Passes today by luck. Sort terms ascending before sum (like `canonical_size` does) so guarantee is structural.

2.3.3 15. Non-finite boxes silently vanish

`score_and_reinsert` drops $\pm\infty$ /NaN without counter. Document + consider counter so NaN storms are visible.

2.3.4 16. Missing crate docs — `crate-level` `//! =`

No top-of-lib blurb. Write what DIRECT is, what's oracle-tested, how it links to C++, what's not implemented ($\mathbb{Q}$, budget bounding).

3 CE Code-Review Synthesis

Reviewers: 6 dispatched, 6 returned, 47 findings, 24 testing gaps, 24 residual risks. Confidence gate ≥ 75 , cross-reviewer promotion applied (`merged` = `raw` = 47). Detail-tier `why` $\neq$ evidence in per-reviewer `output.json`.

3.1 P1 — Critical (must fix)

3.1.1 P1-1. Budget overshoot — guards once per iteration before scoring whole `2 * |POH| batch`

3.1.2 P1-2. `println!` timing on every `run()` through `cdylib` into Qt host `stdout` (4-way agreement)

3.1.3 P1-3. `cargo fmt --check` fails (import order, blank lines)

3.1.4 P1-4. Only budget contract test is #[ignore]d against unfixed defect

3.2 P2 — High (should fix)

3.2.1 P2-5. size() sum order not permutation-invariant (verified)

3.2.2 P2-6. test+=utils compiled into shipped staticlib (panic! + plotters)

3.2.3 P2-7. Pareto!=new_with_strat!=poh.is_empty() untested

3.2.4 P2-8. Non-finite children deleted vs seed retained (sort_cost $\rightarrow \infty$) — orphans

3.2.5 P2-9. Hull NaN-poisoned by $+\infty$ while pareto_front filters

3.2.6 P2-10. Second run() duplicates seed, drifts calls

3.2.7 P2-11. Dead Pareto extension point (zero consumers)

3.2.8 P2-12. Zero docs; RunOutcome::num_iter means evals not iterations

3.2.9 P2-13. No MSRV

3.2.10 P2-14. No [profile.release] LTO/codegen tuning

3.2.11 P2-15. plotters unconditional in prod

3.2.12 P2-16. u32 budget wrap at MAX $\rightarrow$ infinite loop (release)

3.2.13 P2-17. Only restriction lints warned; correctness/suspicious silent

3.2.14 P2-18. FFI batch UniquePtr<Point6D> per eval (heap + crossing)

3.3 P3 — Polish (pick when touched)

3.3.1 P3-1. Dead call_offset field — initialized to 0, never written, baked into budget guard

3.3.2 P3-2. pub everywhere vs pub(crate) — inconsistent with siblings

3.3.3 P3-3. Duplicated denormalize vs test_support::denorm

3.3.4 P3-4. Duplicated helpers show()/zero()/dist() in bench.rs:306

3.3.5 P3-5. Anonymous (Pose, f64) vs named RunOutcome

3.3.6 P3-6. Stale comment determine_potentially_optimal

3.3.7 P3-7. Vendored plotters doc-example test at fixed path

3.3.8 P3-8. #[allow(non_camel_case_types)] vs allow_attributes ban

3.3.9 P3-9. Missing #[must_use] on run() / best()

3.3.10 P3-10. Vec churn per iteration

3.3.11 P3-11. bench.rs naming — fixtures/tests, not a harness

3.3.12 P3-12. run_rust_opt diverges from run() — FFI reports start-position fallback and lifetime-cumulative num_iter

3.3.13 P3-13. size() ULP key splits → phantom size columns

4 How to Fix: test+utils in Shipped Lib

Do not edit — do this yourself:

In rust/direct-rs/src/lib.rs:

```

1  #[cfg(test)]
2  mod bench;
3  #[cfg(test)]
4  mod test;
5  #[cfg(test)]
6  mod utils;
7
8  #[cfg(test)]
9  mod properties;
10 #[cfg(test)]
11 mod test_support;
```

In rust/direct-rs/Cargo.toml:

```

1  [dependencies]
2  cxx = { workspace = true }
3  ordered-float = "5.5.0"
4  # plotters removed
5
6  [dev-dependencies]
7  plotters = { version = "0.3.7", default-features = false, features =
   ↪  ["svg_backend", "line_series", "point_series"] }
```

Verify:

```

1  pixi run cargo check           # no plotters in graph
2  pixi run cargo test --lib      # still 46 passed / 2 ignored
3  cargo tree | grep plotters     # empty on normal build
```

Optional dev-plot feature: `#[cfg(any(test, feature="dev-plot"))]= +plotters = { optional = true }`.

5 Testing Gaps & Residual Risks (aggregated)

5.1 Testing gaps

- Repeated `run()` on same instance (5/6 reviewers)
- FFI surface (`new_rust_opt`, `run_rust_opt`, `Cost` for `CppCost`, `RunOutcome` semantics) uncovered
- Pareto strategy untested
- Non-finite-cost tiling invariants not exercised
- `plots/*.svg` written during `cargo test` without RAI cleanup
- `volume_partition_holds_for_random_runs` silently passes on `Err("pow overflow")`
- `bench.rs` name implies harness but none

5.2 Residual risks

- Silently-dropped `+Inf` subregions orphan domain coverage
- `OrderedFloat(+∞)` vs `NaN` divergence seed children
- `next_box_id`: `u64` exhaustion (theoretical)
- Hostile `CppCost` returning mixed `NaN/±Inf` can poison hull
- `Hyperbox::size()` ULP key splits → phantom size columns

6 Consolidated Prioritized Checklist (do in this order)

1. ☐ `fmt + println!` removal + budget cap (P1 1-3) — one `jj new`; re-enable budget property test
2. ☐ Gate `bench/test/utils` + demote plotters to dev-dependencies (P2 #6 + #15)
3. ☐ `MSRV` + `[profile.release]` (P2 #13-14)
4. ☐ `From<[f64;6]>` for `Pose` + docs on public API (Tier1 #2-3, P2 #12)
5. ☐ Batch FFI `evaluate_batch` (Tier1 #4 / P2 #18) — biggest wall-clock win after #1
6. ☐ `powi` + permutation-invariant `size()` + `Vec` reuse buffers (Tier2 #7-8, P2 #5)
7. ☐ Jones $\boxtimes$ -condition experiment (Tier3 #13)
8. ☐ Promote correctness/suspicious lints + `missing_docs` + `cargo fmt --check` in CI (P3)

7 Verdict

Ready with fixes — land the P1 batch and the crate is clean to widen the seam. P2/P3 are follow-up `jj new` commits. Your Jones oracle + `volume/lattice` + affine-equivariance + prefix-of-budget suite is the right harness — keep it as you add `evaluate_batch` and $\boxtimes$.

8 Appendix: Sources

- Tutoring rubric: `/home/ajj/.pi/agent/skills/rust-skills/SKILL.md` + `rules/*.md` (read per-reviewer: `obs-*`, `err-*`, `num-*`, `api-*`, `mem-*`, `opt-*`, `test-*`, `lint-*`, `proj-*`)

- CE workflow: 052a3f67-213b-4245-8c08-6f7979b1367f, reviewer runs: 945b7640, 591488e8, fee591f6, a2026954, e7a35ede, 9676e07b
- Prior fixes landed: pose/dup_{diag} deletion, ghost Cost removal, #[cfg(test)] cleanup, constructor dedupe, oracle alignment